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Heterogeneous effects of e-government on urban and rural public services: the perspective of the digital divide

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ABSTRACT

This article studies the effects of e-government on both urban and rural public services, to explore its impact on promoting equity in public services. Using individual-level data from the China Family Panel Studies, the empirical results show that e-government raises satisfaction with public services among both urban and rural populations; however, the magnitude of this positive effect is significantly smaller for rural populations. Further analysis reveals that the digital divide is an important factor underlying these disparities. Given that rural populations exhibit less internet access and lower level of digital literacy, improvements in internet access and digital literacy have a more pronounced impact on e-government's role in promoting rural public services than on its role in urban public services. Such improvements enable rural populations to more effectively utilize online government service platforms to access public service information, handle administrative procedures, and communicate with government agencies. Moreover, among rural populations, older individuals and those in less developed regions face greater digital barriers, and consequently benefit less from e-government in terms of public service improvements. These findings suggest that bridging the digital divide is essential to strengthen the role of e-government in enhancing rural public services and promoting equity in public services.

KEYWORDS

E-government; urban and rural public services; digital divide; equity

JEL CLASSIFICATION

H11; H41

I. Introduction

The urban-rural development gap is a prominent issue particularly in developing countries, which is crucial to both efficiency and equity. One important aspect is the disparity between urban and rural public services. Due to geographical distance and disparities in public investment as well as socio-economic development, rural residents typically have less access to public services and receive lower-quality services than urban residents. Though data from the World Bank's Worldwide Exclusion Indicators show that the urban-rural gap in access to public services has gradually narrowed, it remains significant and cannot be ignored. Of the 175 economies surveyed in 2020, 59 had over 75% of their rural population lacking access to good-quality basic public services, and 103 had at least 25% lacking access.[1] Therefore, there is still a great need to improve rural public services.

E-government is seen as a powerful tool to improve the efficiency of public services and public administration by using information and communications technology (ICT). Recently, it has developed rapidly. United Nations (2024) found that 189 of the 193 member states offered at least one online transactional service in 2024. It is also expected that e-government can enhance rural public services and promote equity by providing equitable access to public services and enhancing citizen engagement in public policy making (M. R. Hoque and Sorwar 2015; World Bank 2016).

However, there is a growing concern that because of the digital divide between urban and rural residents, e-government may not necessarily enhance rural public services. The digital divide is the gap between those who have access to and use ICTs including internet connectivity, internet-enabled devices and digital literacy skills and those who do not, and much of the world's digital divide is urban/rural (United Nations Human Settlements Programme 2021). Worldwide, only 50% of rural dwellers used the internet in 2023, while this percentage was 81%

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[1]Data source: Access to public services distributed by urban-rural location, from the World Bank's Worldwide Exclusion Indicators.
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of urban dwellers.[2] This gap is even larger in less developed economies. The digital divide may exacerbate inequalities which is called digital inequalities (Ma 2022). It is an important area of public policy debate (Afshar Ali, Alam, and Taylor 2020). Literature has found that rural residents perceive the use of e-government services as an administrative burden and benefit less from e-government because of inadequate access to the internet and limited digital literacy (Omweri 2024; Reissig, Stoinescu, and Mack 2022).

Therefore, the effect of e-government on rural public services and its effect on promoting equity in public services is still unclear, and there is limited empirical evidence. To enrich the relevant literature, this article empirically compares the effect of e-government on urban and rural public services, and further examines how the digital divide influences e-government's effect on urban and rural public services. Using individual data from the China Family Panel Survey and the performance evaluation of online government service platforms, we find that while e-government improves rural populations' satisfaction with public services, the effect is significantly smaller than that on urban populations. Further analysis reveals that the digital divide may contribute to e-government's heterogeneous effects on improving urban and rural public services, and that improving rural internet access and digital literacy can greatly enhance e-government's positive impact on rural public services, thus enhancing e-government's role in promoting equity in public services.

This article contributes to the literature on both e-government and public services. First, it empirically studies the effects of e-government on both urban and rural public services and explores the impact of the digital divide, which helps to understand the effect of e-government on promoting equity in public services. Many empirical studies have examined the efficiency effect of e-government on country governance and economic development. They find that adopting e-government helps to deliver public services, improve government effectiveness and control corruption (Elbahnasawy 2014, 2021; Kochanova, Hasnain, and Larson 2020; Moreno-Enguix, Lorente-Bayona, and Gras-Gil 2019; Zou et al. 2023), and then has profound impact on economic development, such as promoting economic growth, entrepreneurship and innovation (Almeida and Zouain 2016; Castro and Lopes 2022; Das and Das 2022; Zhu and Yu 2024). Though the equity effect of e-government has also received considerable attention, no consensus has been reached on this impact, and empirical studies remain limited. E-government is perceived to be a useful tool to promote equity and reduce poverty (Lau 2007), and literature has argued that it can improve rural public services (S. M. S. Hoque and Kabir 2024). However, others argue that e-government may not significantly enhance rural public services due to the digital divide (Chary 2011; S. M. S. Hoque 2020; United Nations 2024).

Second, we also provide new insight into rural development in the digital era. The urban-rural development gap is a critical issue of concern to scholars and policymakers. A growing body of literature has examined the impact of digital technology on rural development (Gorla 2009; M. R. Hoque and Sorwar 2015; G. Li and Qin 2022; Thadaboina 2009), as it is considered as an important way to promote rural development. Contributing to this field, this article provides evidence on the effects of e-government on rural public services, offering significant policy implications for improving them through e-government. Our empirical findings also indicate that bridging the digital divide is crucial for adopting digital technologies to promote rural development.

The remainder of this article is organized as follows: Section II presents theoretical analysis. Section III describes the empirical design. Section IV shows the empirical results. Section V concludes.

[2]Data source: ITU World Telecommunication/ICT Indicators database.

II. Institutional background and theoretical analysis

Institutional background

E-government development

E-government is given high priority and has developed rapidly in China. The E-Government Development Index (EGDI) is a well-known index assessed by the United Nations. China's EGDI ranking among 193 member states has risen from 74th in 2003 to 35th in 2024. The number of online government service users grew from 239 million in 2016 to 1.004 billion in 2024, with its share of all internet users rising from 32.7% to 90.6%.[3]

The remarkable progress in e-government development is attributed to strategic policies, innovative initiatives, and investments in digital infrastructure (United Nations 2024). Policies like Internet Plus have integrated internet technologies in traditional industries, enhancing services delivery and public administration. The rapid adoption of artificial intelligence (AI), big data technology, and blockchain has improved efficiency, transparency, and responsiveness in government services. Supportive regulatory frameworks foster innovation while ensuring data security and privacy, building public trust in digital services. Furthermore, investments in high-speed broadband, 5 G networks, and cloud computing ensure seamless connectivity and efficient utilization of digital technology, providing crucial technical support for e-government.

The key and most representative development of e-government in China is the integrated online government service platform. China aims to build an integrated government service platform, in order to achieve service standardization and enhance intergovernmental and interdepartmental collaboration. The most significant feature of the platform is integration and standardization. First, this platform is an online portal that integrates various government services from different departments so as to provide one-stop service for enterprises and the public, which greatly improves administrative efficiency and makes people receive public services with small administrative burden. Specifically, the platform generally provides latest news, information disclosure, online administrative services, and interaction with the government. Online administrative services integrate services from many departments such as the tax bureau, education department and social security department. Thus, people can learn about latest information on different public services, deal with administrative affairs, make inquiries and give feedback on just one online platform, without having to go to multiple departments offline. In 2024, the proportion of provincial-level administrative licencing items that could be processed online and completed within one visit reached 92.5%.[4]

Second, the platform integrates multiple access methods to maximize convenience. Government websites, which were the first and remain the primary means of access, allow the public to receive government services online via computers. In fact, government websites represent a foundational practice of e-government worldwide. With the growing popularity of smartphones and the rise of communication and social platforms such as WeChat, Weibo, and Alipay, online government service platforms have expanded to include access through mini-programs and mobile apps. As a result, access to digital government services is no longer limited to computers and has become more flexible and integrated.

Thirdly, the platform integration spans an increasingly broad range of government levels. Guidance on e-government development is provided for provincial governments. For example, one of the key objectives outlined in the Guiding Opinions of the State Council on Accelerating the Advancement of 'Internet Plus Government Services', issued on 29 September 2016, is to establish integrated online government service platforms in all provincial-level governments by the end of 2017. The online government service platform has now achieved a high degree of integration at

[3]Source: China Internet Network Information Center. 2017. The 39th Statistical Report on China's Internet Development. China Internet Network Information Center. 2025. The 55th Statistical Report on China's Internet Development.
[4]Source: China Internet Network Information Center. 2025. The 55th Statistical Report on China's Internet Development.

the provincial level. Furthermore, it is in the process of integrating at the national level. In this way, it is more convenient for people to handle administrative affairs and access public services across different regions, and the administrative processes are also more standardized.

Digital divide

In the past 30 years, China has achieved remarkable progress in internet popularization. The internet penetration rate increased from 8.5% in 2005 to 78.6% in 2024.[5] Nevertheless, there is a significant urban-rural digital divide. Although this gap has narrowed alongside growing internet access, the disparity remains evident: as of 2024, the urban internet penetration rate stood at 85.3%, compared to only 67.4% in rural areas; non-internet users were still concentrated largely in rural areas, accounting for 54.4% of all non-netizens – a proportion 20.6% points higher than the rural share of total population.[6]

The digital divide is manifested not only by physical access (e.g. an internet connection) but also material access (e.g. digital literacy and usage) (van Deursen and van Dijk 2019). Specifically, a lack of computer/internet literacy and limited internet access devices are two main reasons why non-netizens in China do not use the internet, followed by a lack of need or interest in the internet, and a lack of internet connection.[7] The urban-rural digital divide in China is clearly manifested in the gaps in internet access and digital literacy. For example, in terms of internet access, the number of computers per 100 households was 23.1 in rural areas compared to 58.2 in urban areas as of 2023.[8] Regarding digital literacy, a 2011 survey showed that 57.8% of rural non-internet users cited a lack of computer/internet literacy as the barrier to internet use, a figure significantly higher than that among their urban counterparts (45.7%).[9]

China has made great effort in bridging the digital divide between urban and rural populations. To enhance internet access in rural areas, Broadband Village Project has been implemented since 2014 to expand broadband connectivity and 4 G coverage in rural villages through investments in digital infrastructures. With the Advancement of Digital Rural Initiatives, broadband coverage had been extended to every village by November 2021, and rural areas have been gradually covered by 4 G and 5 G networks. Furthermore, to enhance the affordability of internet devices, the 'Home Appliances Going to the Countryside' policy has been implemented to provide subsidies for rural residents purchasing home appliances. For instance, between 2009 and 2012, a 13% subsidy was offered to rural residents who bought qualifying computers and mobile phones. As for digital literacy, China achieved the universalization of nine-year compulsory education and eliminated illiteracy among young and middle-aged people by November 2011. Significant investments have been made to improve the quality of public education in rural areas, thereby raising the overall educational level of the rural population. Local governments also have provided training on using digital technology for rural residents and offered courses on digital technology at school.

Recently, the urban-rural digital divide in China has been gradually narrowing. Over the past 10 years, the gap in internet penetration rates between urban and rural areas has narrowed significantly, from 34% in 2014 to 17.9% in 2024.[10]

Theoretical analysis

On one hand, e-government can greatly reduce information costs and enforcement costs in rural public service provision. The efficient provision of public goods is the equilibrium between public demand and supply. However, in practice,

[5]Source: China Internet Network Information Center. 2006. The 17th Statistical Report on China's Internet Development. China Internet Network Information Center. 2025. The 55th Statistical Report on China's Internet Development.
[6]Source: China Internet Network Information Center. 2025. The 55th Statistical Report on China's Internet Development.
[7]For example, the 53rd Statistical Report on China's Internet Development shows that in 2023, 51.6% of non-internet users were offline due to a lack of computer/internet literacy, while 16.7% cited a lack of access devices, and 13.6% had no need or interest in using the internet. The 43rd Statistical Report on China's Internet Development shows that in 2018, 4.3% of non-internet users were offline because internet connection was unavailable in their areas.
[8]Source: National Bureau of Statistics.
[9]Source: China Internet Network Information Center. 2012. The 29th Statistical Report on China's Internet Development.
[10]These figures are calculated based on data from the 35th and 55th Statistical Report on China's Internet Development.

information costs and enforcement costs can undermine the efficient provision of public goods. Specifically, technological constraints and low administrative efficiency can increase the cost of public goods provision and reduce supply efficiency. It is also challenging to fully understand public demand for public goods due to information asymmetry. These costs are more pronounced in rural public service provision due to factors such as relatively underdeveloped socio-economic conditions, lower administrative efficiency, and geographical remoteness. These factors also make it more difficult to learn about the public service needs of rural populations.

Utilizing digital technologies, e-government can improve administrative efficiency, especially in rural areas, as administrative affairs can be processed more instantly and in a more standardized manner without being constrained by geographical distance or time limitation. The online government service platform greatly reduces the public's cost of obtaining public service information and dealing with administrative affairs. This is especially important for rural residents, as the online government platform allows them to access public services without having to make lengthy journeys to town administration centres. In this way, the supply cost of rural public services can be reduced. Besides, rural residents can communicate their needs and opinions on public services to the government more conveniently, so that the government may pay more attention to rural public service demand and its improvement. Therefore, e-government can greatly reduce information costs and enforcement costs in rural public service provision, thus significantly enhancing the accessibility and quality of public services in rural areas. S. M. S. Hoque and Kabir (2024) have found that the rural dwellers in Bangladesh are satisfied with the e-government services in terms of their availability, cost, convenience and delivery.

On the other hand, given the urban-rural digital divide, the relatively lower digital literacy and internet access among rural populations may hinder their utilization of e-government services. According to Amartya Sen's Capability Approach, well-being should be understood in terms of people's capabilities, and capabilities are about the freedom and ability to choose a life they have reason to value. Nowadays, digital technologies have been widely adopted in work and daily life. Digital literacy and internet access have become important capabilities to achieve well-being in the digital era. Internet access is the fundamental infrastructure for the use of digital technologies including e-government services. According to UNESCO, digital literacy involves the confident and critical use of a full range of digital technologies for information, communication and basic problem-solving in all aspects of life.[11] Individuals with higher digital literacy are more willing and capable of utilizing digital government services. As a result, the digital divide implies that rural residents lack sufficient abilities and opportunities to use e-government services. They are less likely to learn about public service information, deal with administrative affairs, and put forward their opinions and suggestions via the online government service platform. A survey in 2020 found that 24.9%, 24.6% and 22.9% of non-netizens had difficulty in purchasing tickets/making medical appointments, handling administrative affairs, and accessing news information, respectively.[12] Therefore, rural residents may receive less benefits from e-government, and the effect of e-government on improving rural public services may be smaller than urban public services.

Recently, the urban-rural digital divide in China has gradually narrowed. Improved internet access and digital literacy enable rural populations to adopt digital technologies. This allows them to utilize online government services more effectively, thereby strengthening the impact of e-government on rural public services.

Based on the above analysis, it is uncertain whether e-government has a larger promoting effect on rural public services due to the urban-rural digital divide. Thus, we propose the following hypotheses:

[11]Source: Metadata of SDG Indicator 4.4.2. https://www.uis.unesco.org/sites/default/files/medias/fichiers/2025/07/Metadata-4.4.2.pdf
[12]Source: China Internet Network Information Center. 2021. The 47th Statistical Report on China's Internet Development.

Hypothesis 1a: E-government has a larger effect on improving rural public services than urban public services.

Hypothesis 1b: E-government has a smaller effect on improving rural public services than urban public services.

Hypothesis 2: A narrower digital divide strengthens the impact of e-government on rural public services.

III. Empirical design

Data

This study uses individual-level data from the China Family Panel Studies (CFPS) for empirical analysis. The CFPS is a nationally representative survey of communities, families, and individuals in China, focusing on the economic and non-economic wellbeing. It provides detailed information about the respondent's internet usage and views on public services. It was launched in 2010 by Peking University in China and is conducted every two years. We use data surveyed in 2010, 2012, 2014, 2016, 2018 and 2020, and restrict the sample to respondents aged 16 and above.

Empirical model

First, this article estimates the following basic model with urban and rural samples respectively, to compare the impact of e-government on urban and rural public services. The basic model is as follows:

$$public_{i,t} = \alpha_0 + \alpha_1 egov_{i,t} + X_{i,t}\alpha_2 + \varepsilon_{i,t} \quad (1)$$

where $public_{i,t}$ represents individual i's evaluation of public services in year t; $egov_{i,t}$ denotes the development of e-government in individual i's province; and $X_{i,t}$ is a vector of control variables. Year fixed effects are included to capture shocks in different years. $\varepsilon_{i,t}$ is the disturbance term. Therefore, α_1 is the parameter of interest that captures e-government's effect on public services. We can test Hypothesis 1a and Hypothesis 1b by comparing the estimated effects of e-government on urban and rural public services.

To more precisely compare the effects of e-government on urban and rural public services, we also estimate the following interactive model:

$$public_{i,t} = \beta_0 + \beta_1 egov_{i,t} \times rural_{i,t} + \beta_2 egov_{i,t} + \beta_3 rural_{i,t} + X_{i,t}\beta_4 + \varepsilon_{i,t} \quad (2)$$

Where $rural_{i,t}$ indicates whether an individual is a rural resident, and β_1 captures the differential effect of e-government.

Second, we estimate the following interactive model separately for urban and rural samples to test how the digital divide influences the effects of e-government on urban and rural public services – that is, to examine Hypothesis 2.

$$public_{i,t} = \theta_0 + \theta_1 egov_{i,t} \times digital_{i,t} + \theta_2 egov_{i,t} + \theta_3 digital_{i,t} + X_{i,t}\theta_4 + \varepsilon_{i,t} \quad (3)$$

Where $digital_{i,t}$ indicates internet access or digital literacy. If the coefficient of $egov_{i,t}\times digital_{i,t}$ (denoted by θ_1), is positive and larger in the rural sample, this indicates that internet access and digital literacy are important factors influencing the effect of e-government on public services. This also suggests that improving rural internet access and digital literacy can reduce the difference in e-government's effects between urban and rural public services.

Variable measurement

(1) Public service level (denoted as public). Subjective evaluation of public services is a common measure for assessing their effectiveness, as it reflects individuals' perceived benefits from public services. We measure public service performance using satisfaction ratings for education, healthcare, and social security – three essential public services. The data are based on the severity assessment of these three issues in the CFPS, respectively defined as ordinal variables with values from 0 to 10. For convenience, we set the satisfaction evaluation to a high value when satisfaction with each public service is high. Let 0 be highly dissatisfied, while 10 be highly satisfied. We then sum the satisfaction ratings of these three public services to create a composite public service satisfaction index.

(2) E-government development (denoted as egov). As online government services are the most representative application of e-government, existing studies often employ their evaluation to measure e-government development (Das and Das 2022; Elbahnasawy 2014; Kochanova, Hasnain, and Larson 2020). For example, EGDI is a well-known index assessed by the United Nations, and one of its key components is provision of online services.

Following the literature, this study employed a provincial online government service capability assessment conducted by the National Academy of Governance to measure e-government development. This assessment is a third-party evaluation commissioned by the government, and has been conducted every year since 2015 and widely used to show the development of e-government. Based on the EGDI indicator system, it defines user-oriented evaluation indicators from five dimensions – the completeness of service modes, the coverage of service items, the accuracy of service guidelines, the maturity level of online processing, and the effectiveness of online services – to measure the level of online government services via government websites, mobile terminals, hotlines, and other platforms. Since the satisfaction ratings pertain to public services from the previous year, we use a one-period lag of *egov* in this analysis. Given that detailed scores on provincial evaluation are available from 2015 to 2020, we employ the CFPS data from year 2016, 2018, 2020.

A provincial-level measure of e-government development is used due to data availability and the integrated nature of e-government development. On one hand, although the e-government services an individual actually encounters are at the city or county level, existing evaluations of e-government development only cover a subset of cities and counties. For example, the assessment data of online government services used in this article includes only 32 cities. As a result, there is a data limitation in obtaining comprehensive city- or county-level information on e-government development. On the other hand, as mentioned in Section II, China aims to build an integrated government service platform, in order to achieve service standardization and enhance intergovernmental and interdepartmental collaboration. Integrated online government service platforms had been established in all provincial-level governments by the end of 2017. Therefore, the development of e-governments across cities within the province adheres to unified standards. In addition, people can handle many administrative affairs across different cities through the provincial online government service platform. This justifies the use of provincial-level measure of e-government development.

(3) Digital divide. The urban-rural digital divide is mainly manifested as the disparities in internet access and digital literacy. Given the data availability, we employ whether one uses the internet to measure both internet access and digital literacy, denoted as *usenet*. This indicator not only reflects whether an individual has access to the internet via computers or smartphones, but also demonstrates their ability to use digital devices and software, as well as information and data literacy – both of which are fundamental components of digital literacy, according to the digital literacy framework proposed by UNESCO.[13]

Besides, while digital literacy is commonly perceived as requiring necessary knowledge and skills on digital technologies, Law et al. (2018) hold the view that attitudes are also necessary for a person to have the commitment and motivation to achieve high-level digital literacy. Therefore, they argue that attitudes should be included in the Digital Literacy Global Framework, which serves as the foundation for the United Nations' Sustainable Development Goal (SDG) Indicator 4.4.2. Law et al. (2018) also show that the attitude to use digital technologies has been included in the digital literacy framework in many countries. As a key aspect of a digital mindset, the willingness to actively learn and use digital technologies is one of the indicators used in China to assess digital literacy and skills.[14] Therefore, we use whether an individual regards the internet as an important information channel to measure the

[13]Source: Metadata of SDG Indicator 4.4.2. https://www.uis.unesco.org/sites/default/files/medias/fichiers/2025/07/Metadata-4.4.2.pdf.
[14]Source: Survey Report on National Digital Literacy and Skills Development (2024). https://www.gov.cn/lianbo/bumen/202410/content_6983266.htm.

Table 1. Variable definition.

Variable	Definition
public	Total satisfaction score of education, medical service and social security; Each public service satisfaction score is an integer ranging from 0 to 10, 0=highly dissatisfaction, 10=highly satisfaction
egov	Scores of online government service capabilities in each province
usenet	1=the respondent uses the internet, 0=otherwise
e-attitude	1=the respondent regards the internet as an important information channel, 0=otherwise
rural	1=rural populations, 0=urban populations
gender	1=male, 0=female
age	Age in years
edu	Years of education
income	The logarithm of per capita household net income
pubstress	Provincial fiscal expenditure divided by provincial fiscal revenue
gdpper	The logarithm of provincial GDP per capita in 2020 RMB

Table 2. Descriptive statistics.

Variable	Observation	Mean	Standard deviation	Min	Max
public	83,251	11.619	6.965	0	30
egov	83,251	81.037	7.888	50.440	96.730
usenet	83,251	0.530	0.499	0	1
e-attitude	83,251	0.423	0.494	0	1
rural	83,251	0.730	0.444	0	1
gender	83,251	0.501	0.500	0	1
age	83,251	45.752	16.780	16	98
edu	83,251	7.793	4.964	0	22
income	83,251	9.636	1.049	0	15.549
pubstress	83,251	2.367	0.924	1.122	10.074
gdpper	83,251	10.445	0.429	9.660	11.609

attitude dimension of digital literacy, denoted as *e-attitude*.

(4) Control variables. Control variables are factors that influence individual *i*'s evaluation of public services, including individual and provincial characteristics. Individual characteristics include age, gender, educational attainment, and household income. Provincial characteristics include fiscal stress and GDP per capita, as fiscally stressed provinces may have limited capacity to invest in public services and provinces with higher per capita GDP may have higher requirements for public services.

The detailed variable definitions are presented in Table 1, and descriptive statistics are shown in Table 2.

IV. Empirical results

A comparison of e-government's effects on urban and rural public services

Baseline results

Table 3 reports the OLS estimates of the effects of e-government on urban and rural public service satisfaction, respectively. Columns (1)-(2) show that the estimated coefficients of *egov* are 0.025 for the rural sample and 0.045 for the urban sample, both statistically significant at the 1% level. Besides, column (3) presents that the estimated coefficient of *egov*×*rural* in the interactive model in Eq. (2) is −0.042 at the 1% significant level. These results indicate that while e-government enhances public service satisfaction among both urban and rural populations, the effects are heterogeneous and significantly weaker in rural populations. These heterogeneous effects may undermine e-government's role in promoting equity in public service provision. Therefore, Hypothesis 1b is supported.

We further examine e-government's impact on the specific public services that comprise the overall public service satisfaction indicator. As Table 4 shows, e-government has smaller effects on improving rural populations' satisfaction with education, medical services, and social security than it does on improving urban populations' satisfaction with these public services. These results provide additional evidence regarding the heterogeneous effects of e-government on urban and rural public services.

As for the control variables, Table 3 shows that male and older respondents have higher satisfaction with public services. Older individuals are more likely to benefit from public

Table 3. The effects of e-government on urban and rural public service satisfaction.

sample	(1) *rural*=1	(2) *rural*=0	(3) all
egov	0.025***	0.045***	0.062***
	(0.004)	(0.007)	(0.006)
egov×rural			−0.042***
			(0.007)
rural			3.974***
			(0.562)
gender	0.637***	0.644***	0.650***
	(0.057)	(0.084)	(0.047)
age	0.063***	0.047***	0.058***
	(0.002)	(0.003)	(0.002)
edu	−0.133***	−0.219***	−0.159***
	(0.007)	(0.011)	(0.006)
income	−0.077***	−0.065	−0.086***
	(0.030)	(0.053)	(0.026)
pubstress	−0.189***	−0.344***	−0.221***
	(0.048)	(0.077)	(0.040)
gdpper	−0.237**	0.172	−0.040
	(0.120)	(0.151)	(0.094)
constant	11.423***	7.166***	7.100***
	(1.320)	(1.748)	(1.098)
Year fixed effects	Yes	Yes	Yes
N	60812	22439	83251
R^2	0.058	0.086	0.068

Standard errors are reported in parentheses. ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$.

Table 4. The effects of e-government on urban and rural populations' satisfaction with specific public services.

	Education			Medical services			Social security		
sample	*rural*=1	*rural*=0	all	*rural*=1	*rural*=0	all	*rural*=1	*rural*=0	all
egov	0.010***	0.023***	0.025***	0.008***	0.015***	0.020***	0.007***	0.008***	0.016***
	(0.002)	(0.003)	(0.002)	(0.002)	(0.003)	(0.002)	(0.002)	(0.003)	(0.002)
egov×rural			−0.016***			−0.014***			−0.012***
			(0.003)			(0.003)			(0.003)
N	60812	22439	83251	60812	22439	83251	60812	22439	83251

Control variables are included. Standard errors are reported in parentheses. ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$.

services such as social security and medical services. Highly educated and higher-income respondents are less satisfied with public services, possibly because they have higher expectations for public services. Furthermore, respondents in areas under greater fiscal pressure tend to rate public service performance lower. This is consistent with the view that fiscal constraints may limit resources available for public service improvement. Public service evaluations are also lower in regions with higher per capita GDP. A potential explanation for this finding is that residents in more developed areas are likely to have higher expectations for public services.

The effects of different dimensions of e-government

We further analyse the effects of different e-government dimensions on both urban and rural public service satisfaction. The capabilities of provincial online government services assessed by the National Academy of Governance are evaluated from five dimensions: the completeness of service modes, the coverage of service items, the accuracy of service guidelines, the maturity level of online processing, and the effectiveness of online services. We estimate the impact of these dimensions with Eqs. (1) and (2) respectively, and the results are presented in Table 5.

These five dimensions of e-government all have positive effects on both urban and rural public service satisfaction, and these effects are statistically significant at least at the 5% level. These results demonstrate that individuals can receive better public services by utilizing online government service platforms to access public service information, handle administrative procedures, and communicate with government agencies more efficiently.

Table 5. The impacts of different e-government's dimensions on urban and rural public service satisfaction.

	completeness of service modes			coverage of service items			accuracy of service guidelines		
sample	*rural*=1	*rural*=0	all	*rural*=1	*rural*=0	all	*rural*=1	*rural*=0	all
egov	0.013***	0.041***	0.054***	0.009***	0.011**	0.027***	0.028***	0.034***	0.054***
	(0.003)	(0.006)	(0.005)	(0.003)	(0.005)	(0.005)	(0.004)	(0.007)	(0.006)
egov×*rural*			−0.044***			−0.022***			−0.032***
			(0.006)			(0.005)			(0.006)
N	60812	22439	83251	60812	22439	83251	60812	22439	83251
	maturity level of online processing			effectiveness of online services					
sample	*rural*=1	*rural*=0	all	*rural*=1	*rural*=0	all			
egov	0.016***	0.029***	0.039***	0.010**	0.031***	0.034***			
	(0.003)	(0.004)	(0.004)	(0.005)	(0.007)	(0.007)			
egov×*rural*			−0.026***			−0.025***			
			(0.005)			(0.007)			
N	60812	22439	83251	37371	13862	51233			

Control variables are included. Standard errors are reported in parentheses. ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$.

However, each of these five dimensions has a smaller impact on improving public service satisfaction among rural populations than on that among urban populations. These heterogeneous effects are statistically significant at the 1% level according to estimated coefficients of *egov*×*rural* in the interactive model. These findings suggest that rural populations are less proficient in utilizing e-government services such as obtaining information, handling administrative procedures, and communicating with government agencies to access public services.

Robustness tests

(1) Alternative measures of e-government development. First, we use the ranking of e-government development in each year to measure *egov*. Larger values in magnitude are designated to indicate higher levels of e-government development. Furthermore, we employ the assessment of provincial government portal website performance conducted by Guomai, a professional consulting firm in China, as an alternative measure. This annual assessment has been widely used to measure the e-government development of local governments in the literature (E. Li et al. 2023; Zhang et al. 2024). Given that the survey by Guomai covers the period 2010-2017, the sample has been expanded to include data surveyed in 2010, 2012, 2014, 2016, 2018 and 2020. A one-period lag of *egov* is also used here. Given that the data for this measure are only available up to 2017, we use its 2017 value as a proxy for the year 2019 when matching with the 2020 survey data. Table 6 reports the OLS estimates based on these two alternative

Table 6. The effects of e-government on urban and rural public service satisfaction: robustness tests.

Alternative measure of *egov*						
	Ranking of *egov*			Another assessment of e-government development		
sample	*rural*=1	*rural*=0	all	*rural*=1	*rural*=0	all
egov	0.0284***	0.0480***	0.0607***	0.0219***	0.0487***	0.0421***
	(0.0037)	(0.0059)	(0.0060)	(0.0022)	(0.0035)	(0.0030)
egov×*rural*			−0.0360***			−0.0190***
			(0.0067)			(0.0033)
N	60812	22439	83251	97565	37985	135550
Alternative measure of public service level				Addressing endogeneity		
	Performance of local governments			IV estimation		
sample	*rural*=1	*rural*=0	all	*rural*=1	*rural*=0	all
egov	0.0013***	0.0036***	0.0044***	0.0952***	0.2222***	0.2268***
	(0.0003)	(0.0005)	(0.0005)	(0.0110)	(0.0259)	(0.0167)
egov×*rural*			−0.0033***			−0.1298***
			(0.0005)			(0.0141)
N	60558	22202	82760	60812	22439	83251

Control variables are included. Standard errors are reported in parentheses. ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$.

measures of *egov*. The results are consistent with the baseline findings.

(2) Alternative measure of public service levels. We use respondents' ratings of local government performance in the previous year as an alternative measure of public service levels. This indicator can serve as a general measure for the level of public services. We defined it as a binary variable. A value of 1 indicates that the respondent considered that the local government had made an achievement (either good or some achievement) last year, while a value of 0 indicates otherwise.

The OLS estimation results, reported in Table 6, show that e-government has a smaller effect on improving ratings of local government performance among rural populations than among urban populations. These results provide additional evidence regarding the heterogeneous effects of e-government on urban and rural public service.

(3) Instrumental variable method. We also employ the instrumental variable (IV) approach to further mitigate potential endogeneity concerns. Our instrument is constructed based on the staggered establishment of public data platforms across provinces.

First, providing machine-readable public data to society through public data platforms is an important part of China's e-government development. It shows the government's efforts to integrate and share vast amounts of data across departments. Since 2012, an increasing number of provinces have launched public data platforms.[15] As a component of e-government development, public data platforms affect public services primarily through enhancing e-government performance itself. Individuals do not receive public services directly from these platforms, making their staggered establishment a quasi-natural experiment from the individual's perspective. The constructed IV passes both underidentification and weak identification tests, supporting its validity.

In addition, using policy shocks as instrumental variables when they cannot be directly or fully incorporated as explanatory variables is a well-established practice in the literature (Cornaggia et al. 2022; Jackson, Johnson, and Persico 2016). As reported in Table 6, the IV estimation results yield findings consistent with the baseline results.

The impact of the digital divide

This section examines how the digital divide, a key factor behind the heterogeneous effects of e-government, shapes its impact on public service satisfaction among urban and rural popluations.

As shown in Table 7, in our sample, the proportion of rural respondents using the internet via computers or mobile devices is 48.1%, while this proportion of urban respondents is 66.4%. Moreover, 51.7% of urban respondents regard the internet as an important information channel, compared to only 38.8% of rural respondents, indicating a lower willingness to use the internet among rural populations. This provides evidence that rural populations have less internet access and lower level of digital literacy than urban populations, confirming the urban-rural digital divide.

The impact of the digital divide is estimated using Eq. (3) and presented in Table 8. Regarding both internet access and digital literacy, the estimated coefficients of *egov*×*digital* are positive. However, they are larger and more statistically

Table 7. Internet access and digital literacy among urban and rural residents.

	usenet		*e-attitude*	
sample	*rural*=1	*rural*=0	*rural*=1	*rural*=0
Mean	0.481	0.664	0.388	0.517
N	60,812	22,439	60,812	22,439

[15]Public data platforms were established in 2012 in Beijing and Shanghai; in 2015 in Zhejiang; in 2016 in Guangdong and Guizhou; in 2018 in Henan, Jiangxi, Ningxia, Shandong and Shaanxi; in 2019 in Fujian, Hainan, Jiangsu, Sichuan, Xinjiang and Tianjin; in 2020 in Guangxi, Hubei, Hunan and Qinghai; in 2021 in Anhui, Gansu, Hebei and Chongqing; in 2022 in Liaoning; in 2023 in Shanxi; and in 2024 in Heilongjiang and Neimenggu.

Table 8. The digital divide and e-government's effects on public service satisfaction.

	digital = usenet		*digital = e-attitude*	
sample	*rural*=1	*rural*=0	*rural*=1	*rural*=0
egov×*digital*	0.036***	0.011	0.029***	0.005
	(0.007)	(0.012)	(0.007)	(0.011)
N	60812	22439	60812	22439

Control variables are included. Standard errors are reported in parentheses. ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$.

Table 9. The digital divide and the effects of different e-government's dimensions.

	digital = usenet		*digital = e-attitude*	
sample	*rural*=1	*rural*=0	*rural*=1	*rural*=0
egov = completeness of service modes				
egov×*digital*	0.0388***	0.0201**	0.0357***	0.0127
	(0.0059)	(0.0098)	(0.0061)	(0.0093)
egov = coverage of service items				
egov×*digital*	0.0298***	0.0234***	0.0227***	0.0269***
	(0.0053)	(0.0087)	(0.0055)	(0.0081)
egov = accuracy of service guidelines				
egov×*digital*	0.0364***	0.0146	0.0325***	0.0165
	(0.0062)	(0.0108)	(0.0064)	(0.0101)
egov = maturity level of online processing				
egov×*digital*	0.0100*	−0.0001	0.0070	−0.0110
	(0.0052)	(0.0084)	(0.0055)	(0.0082)
egov = effectiveness of online services				
egov×*digital*	0.0145*	−0.0005	0.0116	−0.0166
	(0.0077)	(0.0129)	(0.0077)	(0.0115)

Control variables are included. Standard errors are reported in parentheses. ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$.

significant in the rural sample. On one hand, these findings indicate that enhancing both internet access and digital literacy (i.e. the ability and willingness to use digital technologies) more significantly strengthens the effect of e-government on improving rural public services. This suggests that bridging the digital divide narrows the disparity in e-government's impact on improving urban and rural public services, thereby confirming Hypothesis 2. On the other hand, the results also provide evidence that the digital divide can explain the heterogeneous effects of e-government on urban and rural public services. Therefore, improving rural internet access and digital literacy empowers rural populations, enabling e-government to more effectively enhance rural public services and promote equity.

We further analyse how reducing the digital divide narrows the disparity in the effects of e-government on improving urban and rural public services, based on the different dimensions of online government service capabilities. As shown in Table 9, improved internet access and digital literacy more significantly enhance the positive effects of most e-government dimensions on rural public service satisfaction. This enhancement is particularly pronounced for the completeness of service modes, the accuracy of service guidelines, the maturity level of online processing, and the effectiveness of online services.

These findings suggest that improved internet access and digital literacy help rural populations to better utilize online government service platforms to access public service information, handle administrative procedures, and communicate with government agencies, enabling them to receive better public services through e-government.

Further analysis of the digital divide among rural populations

This section further investigates how the digital divide manifests and interacts with e-government across socioeconomic dimensions within rural populations.

First, given the substantial regional development disparities in China, we conduct a comparative analysis of the digital divide between rural populations

Table 10. Internet access and digital literacy among rural populations: variations by region and age.

	region		age	
sample	More developed	Less developed	16–44	≥45
usenet	0.508	0.458	0.794	0.200
e-attitude	0.414	0.366	0.617	0.182
N	27991	32821	28774	32038

Table 11. E-government's effects on public service satisfaction among rural populations: variations by region and age.

	region		age	
sample	More developed	Less developed	16–44	≥45
egov	0.038***	0.019***	0.028***	0.019***
	(0.007)	(0.006)	(0.005)	(0.006)
N	27991	32821	28774	32038

Control variables are included. Standard errors are reported in parentheses. ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$.

Table 12. The digital divide and e-government's effects on public service satisfaction among rural populations: variations by region and age.

	region		age	
sample	More developed	Less developed	16–44	≥45
digital=usenet				
egov×digital	0.039***	0.050***	0.028**	0.053***
	(0.011)	(0.010)	(0.012)	(0.014)
digital=e-attitude				
egov×digital	0.027**	0.047***	0.019*	0.037**
	(0.011)	(0.011)	(0.010)	(0.015)
N	27991	32821	28774	32038

Control variables are included. Standard errors are reported in parentheses. ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$.

in more and less developed regions. Based on the median per capita GDP of provinces from the preceding year, provinces in the sample are categorized into two groups: more and less developed regions.

As presented in Table 10, rural respondents in more developed regions show higher proportions of both internet usage and perception of the internet as an important information channel compared to those in less developed regions. This demonstrates their greater internet access and higher digital literacy. Furthermore, Table 11 shows that e-government has a positive effect on public service satisfaction among rural populations in both more and less developed regions, but the effect is smaller in the less developed regions. The estimated coefficient of the interaction term (*egov×digital*) in Table 12 is positive and larger in less developed regions. This suggests that the digital divide helps to explain the heterogeneous effects of e-government on rural public services across more and less developed regions. Therefore, bridging the digital divide enables e-government to enhance rural public services more effectively in less developed regions.

Second, we examine the digital divide by age within rural populations. Older people face significant digital barriers (World Bank 2016), an issue that is especially pronounced in rural areas. As shown in Table 10, only 20% of the rural population aged over 45 uses the internet, a rate significantly lower than that among rural 16- to 44-year-olds (79.4%). Furthermore, only 18.2% of the rural population aged 45 and above regard the internet as an important information channel, compared to 61.7% among those aged 16–44, indicating a lower willingness to use the internet among older rural populations.

Significant digital barriers hinder rural elderly populations from effectively utilizing e-government services. Consequently, as presented in Table 11, e-government has a smaller positive effect on public service satisfaction among rural

populations aged over 45 compared to those aged 16–44. Table 12 further demonstrates that improved internet access and digital literacy significantly enhance the effectiveness of e-government in delivering public services to the rural elderly.

V. Conclusion

Narrowing the urban-rural gap in public services is a critical issue in economic development. E-government is considered a useful tool for promoting equity in public services, but there is a concern that the digital divide could undermine its effectiveness in improving rural public services. Moreover, relevant empirical evidence remains limited. Therefore, using individual-level data from the China Family Panel Studies, this article examines the effects of e-government on both urban and rural public services and further investigates how the digital divide influences e-government's effects.

The results show that e-government enhances both urban and rural public service satisfaction; however, these effects are heterogeneous and significantly smaller among rural populations. This disparity can be explained by rural populations having less internet access and lower digital literacy than urban populations, limiting their ability to utilize online government services. Further analysis shows that improvements in internet access and digital literacy enhance e-government's impact on improving rural public services. These improvements enable rural populations to utilize online government service platforms to access public service information, handle administrative procedures, and communicate with government agencies more effectively. These findings suggest that the digital divide is an important factor underlying e-government's heterogeneous effects on improving urban and rural public services. Therefore, bridging the digital divide enhances e-government's role in promoting equitable public services. We further reveal that among rural populations, older people and those in less developed regions face greater digital barriers, and thus experience smaller public service improvements through e-government.

This study not only provides empirical evidence on e-government's effect in promoting equity in public services, but also has significant policy implications. First, it suggests that e-government can serve as a useful tool for improving rural public services. Digital technology such as AI and big data analysis can be better utilized to facilitate access to public service information, streamline administrative procedures, and enhance citizen-government interactions. The user interface of online government services should also be more user-friendly, with easy operation and improved accessibility. For instance, in addition to web platforms, mobile interfaces should be developed to facilitate access to online government services, such as native smartphone applications and social media-embedded mini-programs. It is also crucial to ensure data security and privacy to increase public trust in digital services.

Second, more measures should be taken to enhance rural internet access and digital literacy, given that the digital divide may weaken e-government's effectiveness in enhancing rural public services and promoting equity in public services. As for internet access, both physical access and material access should be improved. Measures may include (1) enhancing network coverage and connection speeds in rural areas; (2) implementing subsidy programs to cover the costs of digital devices (e.g. computers and mobile phones) and internet service fees for rural populations, thereby making internet access more affordable; and (3) establishing community facilities where residents can access online government services. To improve digital literacy, training programs on using the internet and online government services should be implemented, particularly for rural populations. Improved online government services and internet access also increase their willingness to use e-government services. Besides, more support should be provided to rural elderly populations and residents in less developed regions to enhance their digital literacy and internet access.

These policy implications are particularly instructive for economies that face a pronounced urban-rural digital divide. Policymakers should prioritize improving rural internet access and digital literacy when implementing e-government to enhance public service provision.

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